Tribhuvan University

Faculty of Management

OFFICE OF THE DEAN

Database Management System

BBA — 2nd Semester — 2023 — Answer Sheet

Subject Code: N/A

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

Brief Answer Questions:[10 × 1 = 10]

1. Define Data Model.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Write SQL statement to delete student table.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Differentiate between simple composite attribute.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. List the types of mapping constraints in ER Model.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Write a syntax for check constraint SQL.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Define Transaction.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. What is NoSQL?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Define big data.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. State 1NF.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Why level of abstraction is maintained in database?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Short Answer Questions:(Attempt any FIVE Questions)[5 × 3 = 15]

11. Describe two phase locking mechanism in brief.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. What are the types of database user? Explain each of them in brief.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

13. Explain briefly the importance and methods of Database recovery.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

14. Explain the importance of Creating view with syntax.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

15. What is Database security? How Database is secured? Explain in brief.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

16. What is importance of normalization? Explain with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section C

Long Answer Questions:(Attempt any THREE Questions)[3 × 5 = 15]

17. Identify and explain different types of database languages with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

18. Explain the desirable properties of Transactions.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

19. Explain any five aggregate function used in SQL with syntax.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

20. When generalization and specialization is used in ER model? Support your answer with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section D

Comprehensive Answer / Case / Situation Analysis Questions:[2 × 10 = 20]

21. Employee works on the project under one or more project manager for the company. Client pays for the project that is being developed by the company and client provide the feedback for the project.Draw ER diagram for the above scenario.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

22. Consider following relational database:Account(ac_no, type, balance, branch)Customer(cid, name, address, phone_no, dob)Depositer(ac_no, cid, date)Write SQL statement for the following:i) Insert a tuple {1001, checking, 30000} in Account relation.ii) Delete those accounts which haven't owns any customer.iii) Change the phone_no of Ram to 11111111.iv) Find name of customer who owns an account with balance greater than 2000.v) Display those records of customers whose name starts with "R" and end with "A".

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.